INCH-POUND

MIL-DTL-17/189C
12 June 2014
SUPERSEDING
MIL-C-17/189B
20 February 1991

DETAIL SPECIFICATION SHEET

# CABLE, RADIO FREQUENCY, FLEXIBLE COAXIAL, 50 OHMS, M17/189-00001 UNARMORED, M17/189-00002 ARMORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

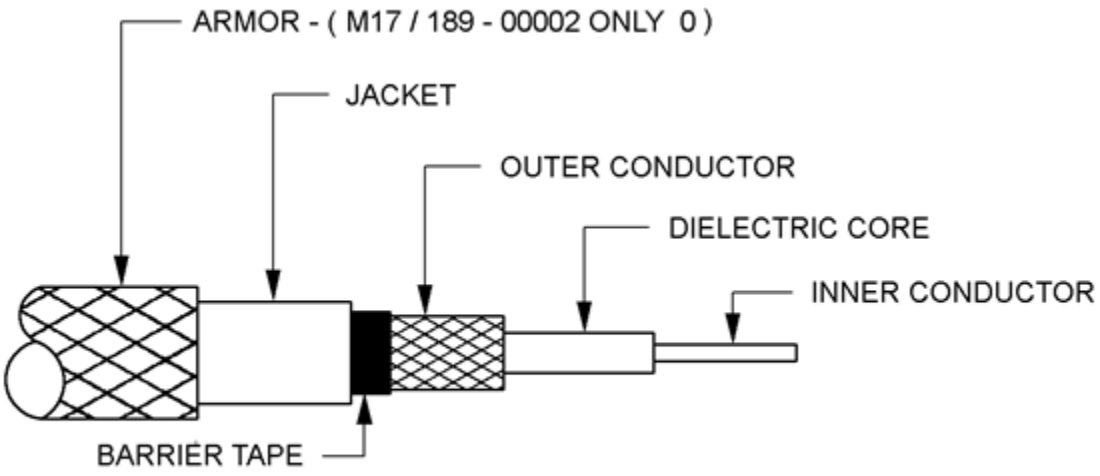



FIGURE 1. Configuration.



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TABLE I. Description.

| Components | Construction details |
|---|---|
| Inner conductor | Seven strands of bare copper wire, each strand .0296 inch diameter. Overall diameter: .0888 inch ± .0010. |
| Dielectric core | Type A-1: Solid polyethylene. Diameter: .285 inch ± .007. |
| Outer conductor | Single braid of AWG No. 33, bare copper wire. Diameter: 0.330 inch maximum.<br><br>Coverage: 95.3% nominal<br>Carriers: 24<br>Ends: 8<br>Picks/inch: 6.5 ± 10% |
| Barrier tape | A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, aluminum face toward the outer conductor. Diameter: .340 inch maximum. |
| Jacket | Cross-linked polyolefin. Diameter: .405 inch ± .007. |
| Armor (M17/189-00002) | Single braid of aluminum-alloy wire. Diameter: .475 inch maximum. |

REQUIREMENTS:

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -30°C to +85°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 0.176 ohms per 100 feet.

Elongation: 25 percent, minimum.

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: 10 percent maximum.

Adhesion of conductors:

Inner conductor to core: 7 pounds, minimum; 50 pounds, maximum.

Aging stability: +98°C ± 2°C.

Cold bend: -30°C ± 2°C.

Dimensional stability: +85°C ± 2°C.



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Inner conductor from core: .062 inch, maximum.

Inner conductor from jacket: .125 inch, maximum.

Contamination: Not applicable.

Flame propagation: Applicable.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50

Elongation, percent of unaged minimum: 50

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation, 160 percent, minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60

Elongation, percent minimum: 60

Hot oil immersion:

Tensile strength, percent minimum: 50

Elongation, percent minimum: 50



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Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight:

M17/189-00001: 12.1 pounds per 100 feet maximum.

M17/189-00002: 14.6 pounds per 100 feet maximum.



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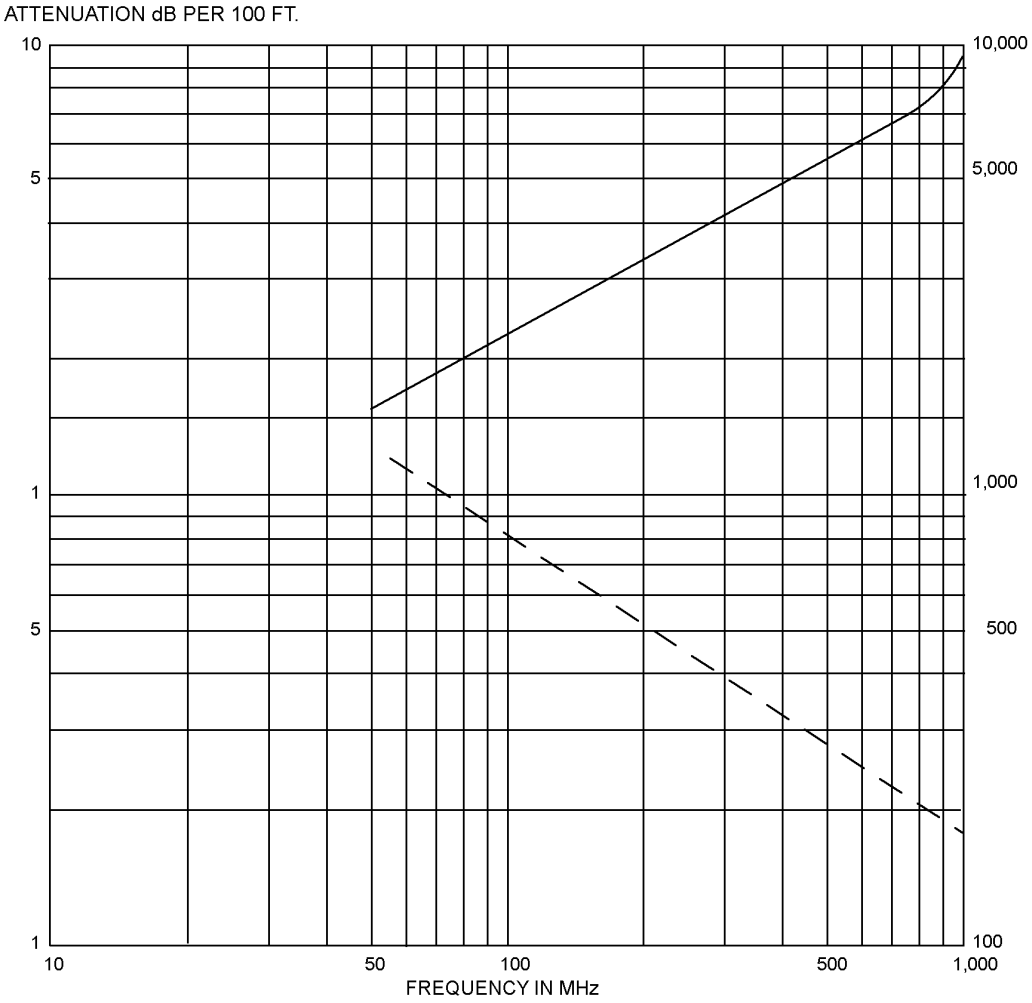



FIGURE 2. Power rating and attenuation.



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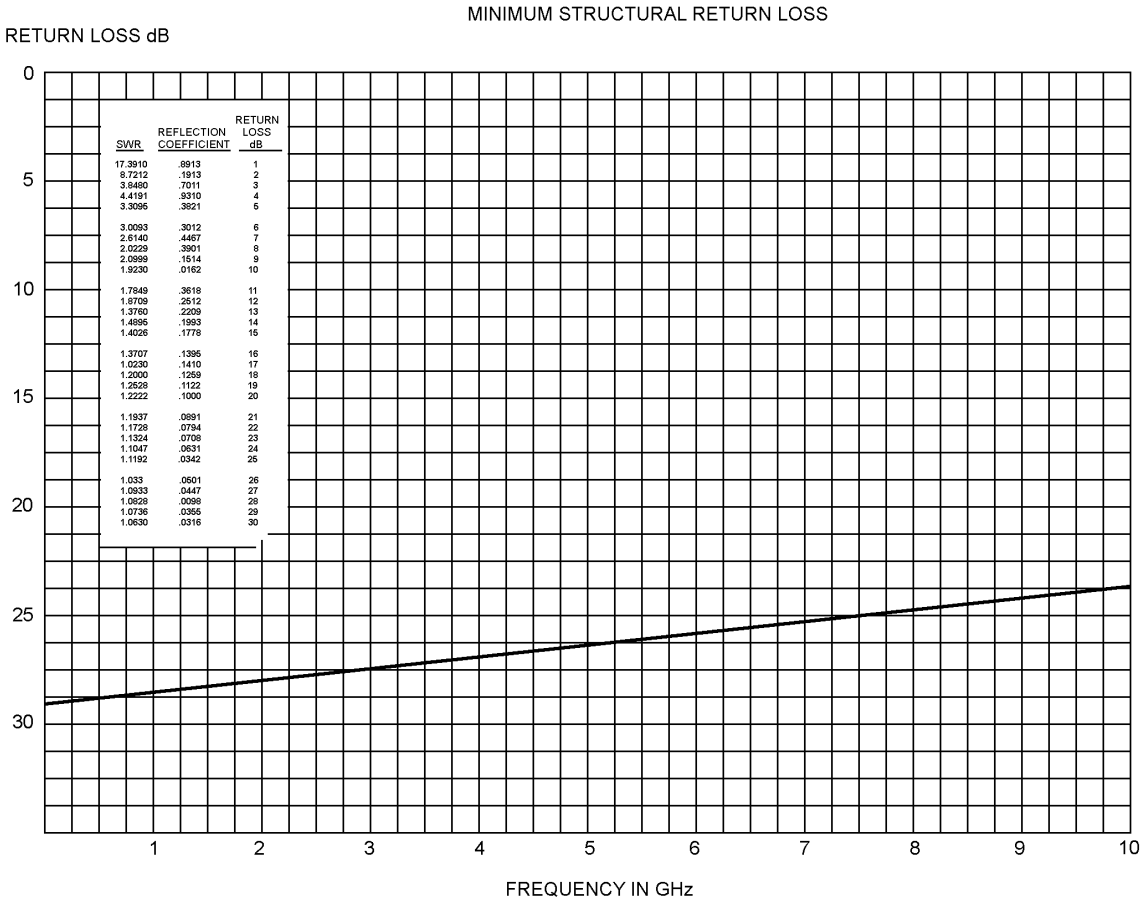



FIGURE 3. Structural return loss.



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Electrical:

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 10,000 V rms, minimum.

Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 50 ± 2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Part or Identifying Number (PIN):

M17/189-00001 unarmored.

M17/189-00002 armored.

NOTE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

(Project 6145-2014-019)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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